Lab 5: Requirement Elicitation - II

System Specification: LOANZO (Peer-to-Peer Microfinance and Purpose-Bound Social Lending Platform)

Activities Conducted:

1. Refinement of Functional and Non-Functional Requirements
2. Representation of Requirements Using Decision Trees
3. Representation of Requirements Using Decision Tables

Document Outcome:

Structured Requirements Document (SRD)

1. Refinement of Functional and Non-Functional Requirements

High-level requirements elicited in Lab 4 are refined into granular engineering specifications with explicit triggers, business rules, operational constraints, and measurable quality thresholds.

1.1 Refined Functional Specifications

Identifier	Refined Requirement Domain	Pre-Condition, Detailed Business Logic & Acceptance Criteria
RFR-01	**Purpose-Bound Tranche* Disbursal**	**Trigger:** Borrower initiates milestone disbursement request. **Logic:** The system extracts vendor UPI VPA* from the commercial tax invoice and executes an automated INR 1.00 penny-drop check*. If beneficiary name matches the invoice (match >= 85%), funds transfer directly to the vendor VPA via NPCI rails. Zero capital is released as unmonitored cash. **Acceptance Criteria:** 100% of disbursed capital must settle directly into verified vendor accounts.
RFR-02	**Multi-Asset Collateral Valuation**	**Trigger:** Borrower submits asset pledge documentation. **Logic:** The system categorizes collateral into 5 classes and enforces statutory Loan-to-Value (LTV)* caps: Gold Bullion (75% IBJA benchmark), Property Deeds (55%), Motor Vehicles (60% Form 34 hypothecated*), Commercial IT Equipment (45%), and Accounts Receivable (75%). Asset valuations exceeding INR 100,000 mandate dual-agent independent inspection. **Acceptance Criteria:** Loan requests exceeding statutory LTV ceilings are programmatically blocked.
RFR-03	**Statutory Promissory Note* Synthesis**	**Trigger:** Counterparty bid accepted in marketplace. **Logic:** The system synthesizes an unconditional Promissory Note compliant with Section 4 of the Negotiable Instruments Act, 1881*, accompanied by a Key Fact Statement (KFS). Documents are executed via Aadhaar OTP e-Sign and sealed with a SHA-256 hash, GPS coordinates, and timestamp for legal admissibility under Section 65B Indian Evidence Act. **Acceptance Criteria:** Contract generation must complete in <= 2.5 seconds on-device with zero tamper vulnerability.
RFR-04	**Non-Compounding Regulatory Penalties***	**Trigger:** Installment overdue status detected. **Logic:** The system provides a 3-day grace period with zero penalty. From Day 4, daily penal charges accrue strictly on overdue principal, capped at a maximum 2.0% monthly ceiling per regulatory guidelines (RBI/2023-24/53). Penal charges are tracked separately and never capitalized into loan principal. **Acceptance Criteria:** Cumulative monthly penalty must never exceed 2% of overdue principal balance.

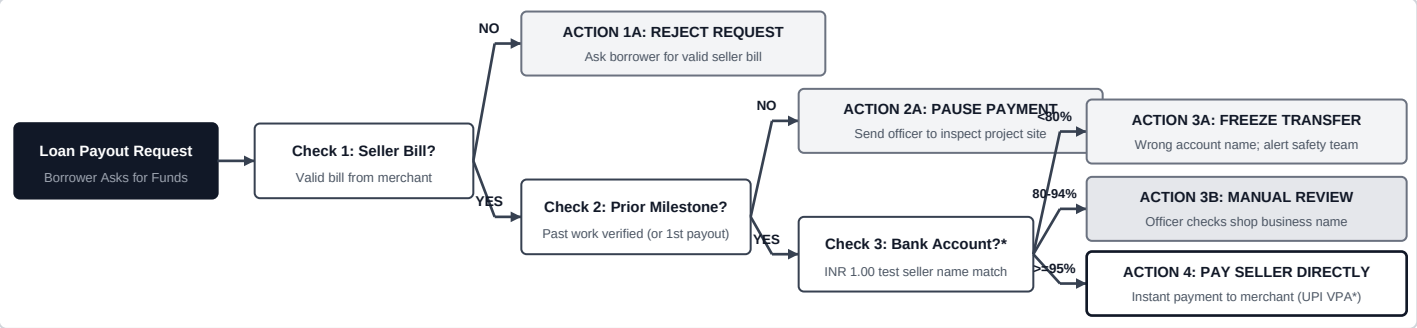
1.2 Refined Non-Functional Specifications

Identifier	Quality Attribute	Operational Requirement & Quantitative Threshold	Verification Method
RNFR-01	**Database Performance**	Local database query execution shall not exceed 15ms (95th percentile) on baseline 2GB RAM devices.	Automated Profiler Suite
RNFR-02	**Application Cold Start**	Launch to interactive Home Dashboard state shall complete in <= 1.8 seconds with zero UI flicker.	Macrobenchmark Test
RNFR-03	**Data Confidentiality**	All stored KYC credentials and documents shall be encrypted at rest using AES-256-GCM via hardware KeyStore.	Static / Memory Heap Audit
RNFR-04	**Offline Fault Tolerance**	Core features (view loans, calculate EMIs, draft agreements) shall execute with 0% network connectivity.	Signal Disconnection Test
RNFR-05	**Atomic Synchronization**	Offline mutations must reconcile within 30 seconds of network restoration via transactional WorkManager queues.	Network Recovery Test
RNFR-06	**Regulatory Non-Custodial***	Platform shall maintain exactly INR 0.00 custodial deposit balance; 100% direct account settlements.	Financial Ledger Audit

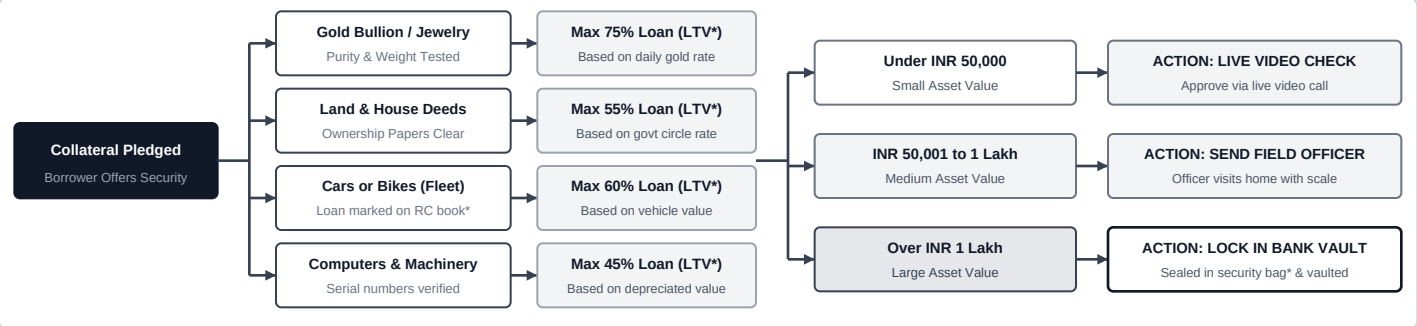
2. Representation of Requirements Using Decision Trees

Decision trees define transparent, automated logic rules for the platform. Every loan disbursement, collateral pledge, or overdue payment follows clear, predictable steps so borrowers, lenders, and administrators always understand the exact system outcome.

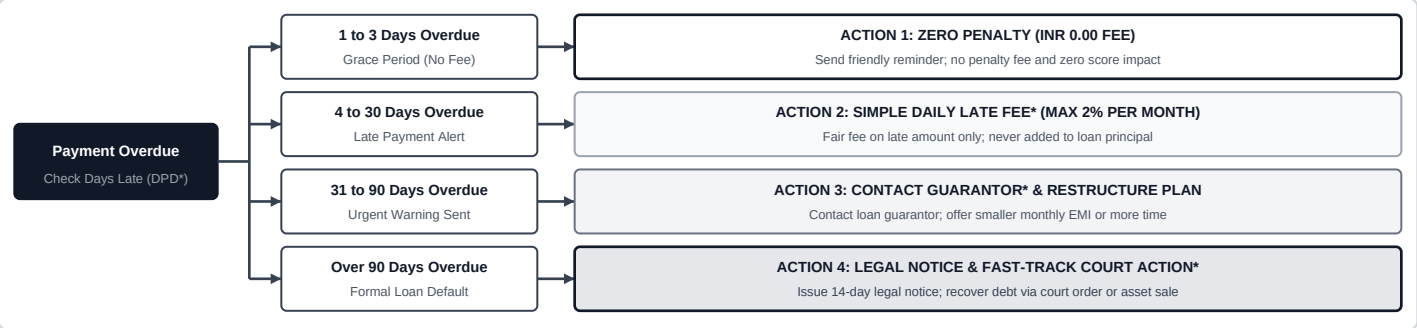
Decision Tree 1: Safe Loan Payout (Milestone Tranches*) and Seller Verification Policy



Decision Tree 2: Collateral Security Valuation, Safety Limits (LTV*) and Inspection Dispatch



Decision Tree 3: Overdue Payments, Late Fees* and Legal Debt Recovery Workflows



* Key Terminology & Plain-English Definitions for General Readers:

* Term	Plain-English Definition & Real-World Meaning
Tranche Disbursement	Releasing a loan in smaller portions/milestones (e.g., Milestone 1 for tools, Milestone 2 for materials) instead of giving the entire cash upfront, preventing fund misuse.
Penny-Drop Check	Sending a tiny test payment of INR 1.00 to a vendor's bank account to automatically receive and verify their real, registered account name before transferring loan funds.
UPI VPA	Virtual Payment Address (e.g., *dealer@bank*) -- a unique UPI identifier used to route digital money transfers directly to verified merchants without exposing raw bank details.
LTV Ratio (Loan-to-Value)	The maximum loan percentage allowed against an asset's market value (e.g., 75% LTV on INR 100,000 gold means the borrower receives a maximum loan of INR 75,000).
Haircut (Collateral Discount)	A safety discount deducted from collateral value to protect lenders from market drops (e.g., a 25% safety haircut leaves a 75% loan ceiling).
Form 34 Hypothecation	A formal legal endorsement recorded on a vehicle's official registration certificate (RC Book) noting that the vehicle acts as collateral for the loan.
Promissory Note (Sec 4 NI Act)	A legally binding document signed by the borrower containing an unconditional written promise to repay a specific sum of money to the lender on a specified date.
Simple Late Fee (Non-Compounding)	A flat daily fee on overdue installments that can never be added into loan principal, preventing lenders from charging 'interest-on-interest' (debt traps).
DPD (Days Past Due)	The exact count of days that have passed since an EMI payment was due (e.g., DPD = 5 means the payment is 5 days overdue).
Order XXXVII (Order 37 CPC)	A special fast-track civil court lawsuit in India for debt recovery based on written promissory notes, resolving cases in ~60 days without long 7-year trials.
Guarantor / Co-Signer	A trusted family member or peer who agrees to help repay the loan if the primary borrower faces hardship or default.
Non-Custodial Model	The platform holds INR 0.00 customer deposits; all money transfers move directly from user-to-user or user-to-merchant, eliminating banking license burdens.

3. Representation of Requirements Using Decision Tables

Decision tables provide exhaustive verification of multivariate logic, mapping combinations of input conditions to discrete system actions with zero ambiguity.

Decision Table 1: Loan Application Underwriting and Marketplace Bidding Gate

Element	Condition / Action Parameter	R1	R2	R3	R4	R5
C1	DigiLocker Identity Verified (Aadhaar and PAN)?	Y	Y	Y	Y	N
C2	On-Device Facial Liveness Verification Passed?	Y	Y	Y	N	--
C3	Historical Unresolved Default on Record?	N	N	Y	--	--
C4	Requested Borrowing APR Within Statutory Ceiling?	Y	N	--	--	--
A1	**Publish Loan Requisition to Social Bidding Marketplace**	**X**				
A2	**Prompt Mandatory APR Adjustment to Comply with Cap**		**X**			
A3	**Reject Origination Request and Restrict Lending Privileges**			**X**		
A4	**Prompt Secondary Facial Liveness Verification Challenge**				**X**	
A5	**Halt Origination Workflow; Redirect to KYC Verification Portal**					**X**

Decision Table 2: Purpose-Bound Milestone Tranche Disbursement Verification Gate

Element	Condition / Action Parameter	R1	R2	R3	R4
C1	Commercial Tax Invoice Uploaded by Borrower?	Y	Y	Y	N
C2	Penny-Drop Beneficiary Name Matches Invoice (>= 85%)?	Y	Y	N	--
C3	Prior Milestone Physical Utilization Verified by Field Agent?	Y	N	--	--
A1	**Execute Direct Tranche Settlement to Vendor UPI VPA**	**X**			
A2	**Hold Tranche; Dispatch Field Agent for Physical Audit**		**X**		
A3	**Lock Tranche; Log Security Incident for Compliance Review**			**X**	
A4	**Prompt User to Submit Commercial Invoice Documentation**				**X**

Decision Table 3: Collateral Valuation, Haircut Enforcement, and Field Audit Routing

Element	Condition / Action Parameter	R1	R2	R3	R4	R5
C1	Pledged Collateral Asset Category	Gold	Property	Vehicle	IT Assets	Any
C2	Primary Ownership Documentation Formally Verified?	Y	Y	Y	Y	N
C3	Declared Asset Valuation Exceeds INR 100,000?	N	Y	N	Y	--
A1	**Approve Collateral with Statutory 75% LTV Cap (IBJA)**	**X**				
A2	**Approve Collateral with 55% LTV Cap (Non-Encumbrance)**		**X**			
A3	**Approve Collateral with 60% LTV Cap (Form 34 Hypothecation)**			**X**		
A4	**Approve Collateral with 45% LTV Cap (Serial Checked)**				**X**	
A5	**Mandate Dual-Agent Independent Inspection Protocol**		**X**		**X**	
A6	**Reject Collateral Pledge; Request Ownership Documents**					**X**

4. Outcome: Structured Requirements Document (SRD)

The Structured Requirements Document compiles refined functional and quality specifications into a normalized traceability matrix ready for architectural design and test verification.

Identifier	Requirement Title	Pre-Condition / Trigger	Operational Specification	System State	Verification
SRD-01	**Multi-Factor Authentication**	User initiates application launch	Enforce OTP, password credentials, and BiometricPrompt hardware check.	**AUTHENTICATED**	Security Test
SRD-02	**DigiLocker Identity Ingestion**	Applicant registration triggered	Query DigiLocker for Aadhaar/PAN; run ML Kit facial liveness detection.	**VERIFIED_ID**	Integration Test
SRD-03	**Cryptographic Data Vault**	Sensitive document stored locally	Encrypt artifacts via AES-256-GCM backed by hardware Android KeyStore.	**VAULT_SECURED**	Static Code Audit
SRD-04	**Reverse-Auction Marketplace**	Loan requisition published	Verified retail lenders submit competitive APR and tenure proposals.	**BIDS_ACTIVE**	Functional Test
SRD-05	**Purpose-Bound Tranche Escrow**	Disbursement milestone reached	Route capital directly to verified vendor UPI VPA; bypass borrower.	**SETTLED_VENDOR**	End-to-End API
SRD-06	**Penny-Drop Account Match**	Vendor account submitted	Execute automated INR 1.00 validation matching commercial invoice name.	**NAME_VERIFIED**	API Mocking Test
SRD-07	**Statutory Promissory Note**	Terms accepted by counterparties	Synthesize 4-page Section 4 NI Act promissory note with Key Fact Statement.	**NOTE_SYNTHESIZED**	Document Parser
SRD-08	**Aadhaar Electronic Signature**	Promissory note generated	Execute Aadhaar OTP e-Sign; embed SHA-256 tamper-evident digital digest.	**CONTRACT_SIGNED**	API Audit Test
SRD-09	**Multi-Asset Collateral Escrow**	Asset pledged by borrower	Enforce statutory LTV ceilings across Gold (75%), Property (55%), etc.	**COLLATERAL_LOCKED**	Mathematical Unit
SRD-10	**Dual-Agent Consensus Protocol**	Valuation exceeds INR 100,000	Require independent inspection reports from two certified ground officers.	**DUAL_VERIFIED**	Workflow Test
SRD-11	**Non-Compounding Overdue Fee**	Installment overdue > 3 days	Apply daily fee strictly capped at statutory 2.0% monthly ceiling.	**PENALTY_ACCRUED**	Unit Test Suite
SRD-12	**Distress Restructuring & NOC**	Borrower distress verified	Restructure amortization schedule; issue digital No Objection Certificate.	**RESTRUCTURED**	State Regression